B.Tech. Degree I Semester Regular Examination November 2023

EE/EC/CS/IT 23-200-0105B COMPUTER PROGRAMMING

(2023 *Scheme*)

Time: 3 Hours | Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Identify main components of a computer system and explain its working.
CO2: Develop flow chart and algorithms for computational problems.
CO3: Write the syntax of various constructs of C language.
CO4: Build efficient programs by choosing appropriate decision-making statements, loops and data structures.
CO5: Design modular programs using functions for larger problems.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(5 \times 2 = 10)$	Marks	BL	CO	PO
I.	(a)	Explain the various categories of software.	2	L1	1	1,2,3
	(b)	Is it advisable to use **goto** statements in a C program? Justify your answer.	2	L2	3	1,2,3
	(c)	What is the output of following program? Explain the answer.	2	L3	4	1,2,3

```
#include < stdio.h >
int main () {
int arr [] = {1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip = arr+4;
printf ("%d\n", ip[1]);
return 0;
}
```

			Marks	BL	CO	PO
	(d)	Explain dynamic memory allocation in C.	2	L2	3	1,2,3
	(e)	What is the output of following program? Explain the answer.	2	L3	4	1,2,3

```
#include <stdio.h>
int main()
{ enum days{ mon =-1,tues,we=4,thu,fri,sat};
Printf("%d,%d,%d,%d,%d,%d",mon,tue,wed,thu,fri,sat);
return 0;
}
```

PART B

$(4 \times 10 = 40)$

			Marks	BL	CO	PO
II.	(a)	Explain how linker, loader and interpreter contribute to the overall process of program execution?	5	L2	1	1,2,3
	(b)	Explain the concept of variables and how data types are associated with them. In the following programming scenarios recommend an appropriate data type for a variable based on nature of data it will store. (i) Mark (ii) Temperature (iii) Employee name (iv) Student details	5	L3	2	1,2,3

OR

(P.T.O.)

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			Marks	BL	CO	PO
III.	(a)	Write an algorithm and draw flowchart for the given sequence: Input purchase amount, if purchase amount greater than 10000 then give discount of 5% else 2%, calculate and display discount and net amount.	5	L6	2	1,2,3
	(b)	What are the primary components of a digital computer system and how do they interact to perform computations?	5	L6	1	1,2,3
IV.	(a)	Provide a detailed explanation of entry-controlled and exit-controlled loops, along with appropriate example.	5	L2	3	1,2,3
	(b)	Write a program to display the following output 5 4 4 3 3 3 2 2 2 2 1 1 1 1 1	5	L6	4	1,2,3
		OR				
V.	(a)	Design and develop a C program to reverse an integer number NUM and check whether it is PALINDROME or NOT.	6	L6	4	1,2,3
	(b)	Is there any difference between the pre-decrement and post decrement operators? Explain with suitable examples.	4	L3	3	1,2,3
VI.	(a)	Write a program that reads a sentence and print the frequency of each vowels and total count of consonants.	6	L6	4	1,2,3
	(b)	What is recursion? Write a program in C to find out factorial of a given number using recursion.	4	L2	5	1,2,3
		OR				
VII.	(a)	Write a C program that implements string copy operation that copies a string str1 to another string str2 without using library function.	4	L6	4	1,2,3
	(b)	Explain the concept of function parameter and their role in passing data to a function. Discuss the difference between call by value and call by reference.	6	L2	5	1,2,3
VIII.	(a)	Write a program in C to sort an array using a pointer.	7	L6	5	1,2,3
	(b)	Illustrate difference between structure and union.	3	L2	3	1,2,3
		OR				
IX.	(a)	Write a short note on pointer arithmetic.	5	L1	3	2,3
	(b)	Write a program that allows the user to input details for multiple employees including their salary, and calculate the average salary (using structure).	5	L6	5	2,3

Bloom's Taxonomy Levels
L1 = 10%, L2 = 38%, L3 = 19%, L6 = 33%.

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